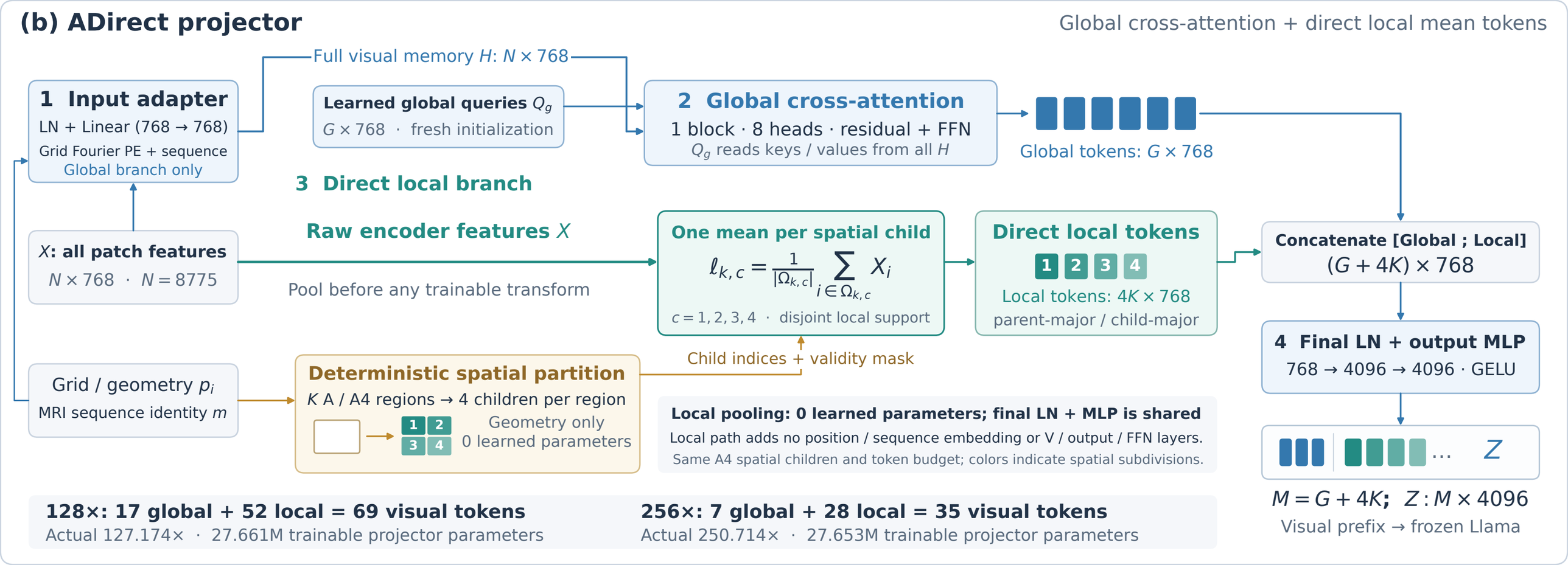
(b) ADirect projector
Global cross-attention + direct local mean tokens
Full visual memory H: N × 768
1 Input adapter
LN + Linear (768 → 768)
Grid Fourier PE + sequence
Global branch only
Learned global queries Q_g
G × 768 · fresh initialization
2 Global cross-attention
1 block · 8 heads · residual + FFN
Q_g reads keys / values from all H
Global tokens: G × 768
3 Direct local branch
Raw encoder features X
X: all patch features
N × 768 · N = 8775
Pool before any trainable transform
One mean per spatial child
$\ell_{k,c} = \frac{1}{|\Omega_{k,c}|} \sum_{i \in \Omega_{k,c}} X_i$
c = 1, 2, 3, 4 · disjoint local support
Direct local tokens
1 2 3 4
Local tokens: 4K × 768
parent-major / child-major
Concatenate [Global ; Local]
(G + 4K) × 768
4 Final LN + output MLP
768 → 4096 → 4096 · GELU
Child indices + validity mask
Grid / geometry p_i
MRI sequence identity m
Deterministic spatial partition
K A / A4 regions → 4 children per region
1 2
3 4
Geometry only
0 learned parameters
Local pooling: 0 learned parameters; final LN + MLP is shared
Local path adds no position / sequence embedding or V / output / FFN layers.
Same A4 spatial children and token budget; colors indicate spatial subdivisions.
Z
$M = G + 4K$; $Z : M \times 4096$
Visual prefix → frozen Llama
128×: 17 global + 52 local = 69 visual tokens
Actual 127.174× · 27.661M trainable projector parameters
256×: 7 global + 28 local = 35 visual tokens
Actual 250.714× · 27.653M trainable projector parameters